

Chapter: Introduction

Ahonakpon Guy Gbaguidi

Exercise 1.12

Pattern Recognition and Machine Learning - Christopher M. Bishop

Solution

In this exercise, we are asked to show that

$$\mathbb{E}[x_n x_m] = \mu^2 + I_{nm} \sigma^2$$

where x_n and x_m denote data points sampled from a Gaussian distribution with mean μ and variance σ^2 , and I_{nm} satisfies $I_{nm} = 1$ if $n = m$ and $I_{nm} = 0$ otherwise.

To prove this, we start by considering the definition of the expectation of the product of two random variables:

$$\mathbb{E}[x_n x_m] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_n x_m p(x_n, x_m) dx_n dx_m$$

We know that if $n \neq m$, then x_n and x_m are independent, and thus the joint probability density function can be expressed as the product of their individual densities:

$$p(x_n, x_m) = p(x_n)p(x_m)$$

Then, we can write:

$$\begin{aligned} \mathbb{E}[x_n x_m] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_n x_m p(x_n) p(x_m) dx_n dx_m \\ &= \int_{-\infty}^{\infty} x_n \left(\int_{-\infty}^{\infty} x_m p(x_m) dx_m \right) p(x_n) dx_n \\ &= \int_{-\infty}^{\infty} x_n \mathbb{E}[x_m] p(x_n) dx_n \\ &= \mathbb{E}[x_m] \int_{-\infty}^{\infty} x_n p(x_n) dx_n \\ &= \mathbb{E}[x_m] \mathbb{E}[x_n] \\ &= \mathbb{E}[x_n] \mathbb{E}[x_m] \\ &= \mu^2 \end{aligned}$$

Now, if $n = m$, then we have:

$$\begin{aligned} \mathbb{E}[x_n x_n] &= \mathbb{E}[x_n^2] \\ &= \text{Var}(x_n) + (\mathbb{E}[x_n])^2 \\ &= \sigma^2 + \mu^2 \end{aligned}$$

We can conclude that:

$$\mathbb{E}[x_n x_m] = \mu^2 + I_{nm} \sigma^2$$

where $I_{nm} = 1$ if $n = m$ and $I_{nm} = 0$ otherwise, which completes the proof.

To continue, we have to prove results (1.57) and (1.58) in the book. First, let's recall that the maximum likelihood solutions μ_{ML} and σ_{ML}^2 are functions of the data points $x_1, x_2, \dots, x_N$. By considering the expectations of these solutions with respect to the data set values, which are drawn from a gaussian distribution with mean μ and variance σ^2 .

we can prove (1.57) and (1.58) as follows:

$$\begin{aligned} \mathbb{E}[\mu_{ML}] &= \mathbb{E}\left[\frac{1}{N} \sum_{n=1}^N x_n\right] \quad (\text{definition of } \mu_{ML}) \\ &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[x_n] \quad (\text{linearity of expectation}) \\ &= \frac{1}{N} \sum_{n=1}^N \mu \quad (\text{each } x_n \text{ has mean } \mu) \\ &= \mu \quad (\text{there are } N \text{ identical terms}) \end{aligned}$$

$$\begin{aligned} \mathbb{E}[\sigma_{ML}^2] &= \mathbb{E}\left[\frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})^2\right] \quad (\text{definition of } \sigma_{ML}^2) \\ &= \frac{1}{N} \sum_{n=1}^N \mathbb{E}[(x_n - \mu_{ML})^2] \quad (\text{linearity of expectation}) \\ &= \frac{1}{N} \sum_{n=1}^N (\mathbb{E}[x_n^2] - 2\mathbb{E}[x_n \mu_{ML}] + \mathbb{E}[\mu_{ML}^2]) \quad (\text{expand the square}) \\ &= \frac{1}{N} \sum_{n=1}^N \left(\sigma^2 + \mu^2 - 2\left(\mu^2 + \frac{\sigma^2}{N}\right) + \left(\mu^2 + \frac{\sigma^2}{N}\right) \right) \\ &\quad (\text{use } \mathbb{E}[x_n^2] = \sigma^2 + \mu^2, \mathbb{E}[x_n \mu_{ML}] = \mu^2 + \sigma^2/N, \text{ and } \mathbb{E}[\mu_{ML}^2] = \mu^2 + \sigma^2/N) \\ &= \frac{1}{N} \sum_{n=1}^N \sigma^2 \left(1 - \frac{1}{N}\right) \quad (\text{simplify each summand}) \\ &= \frac{N-1}{N} \sigma^2 \quad (\text{sum } N \text{ identical terms}) \end{aligned}$$

Q.E.D.